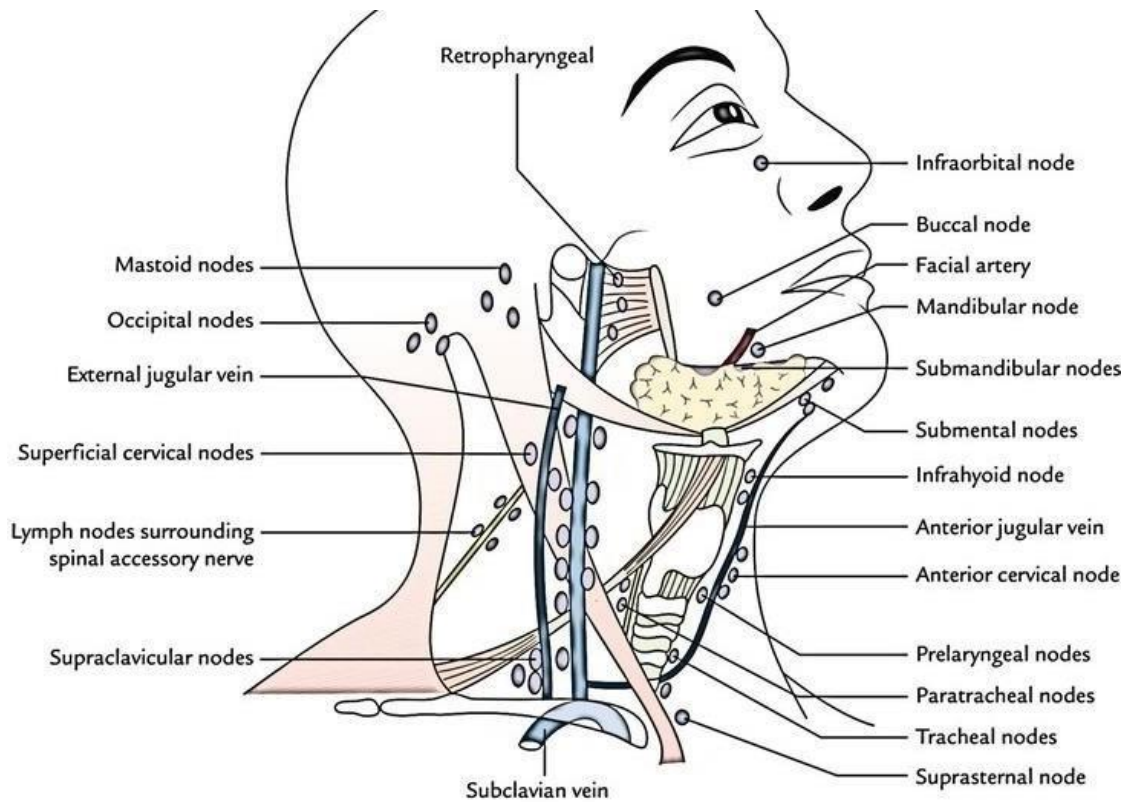
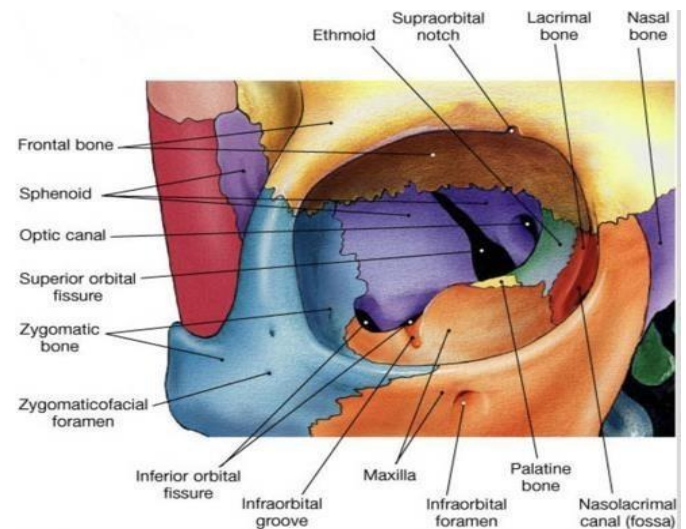




Lecture (6): Contents of the orbit

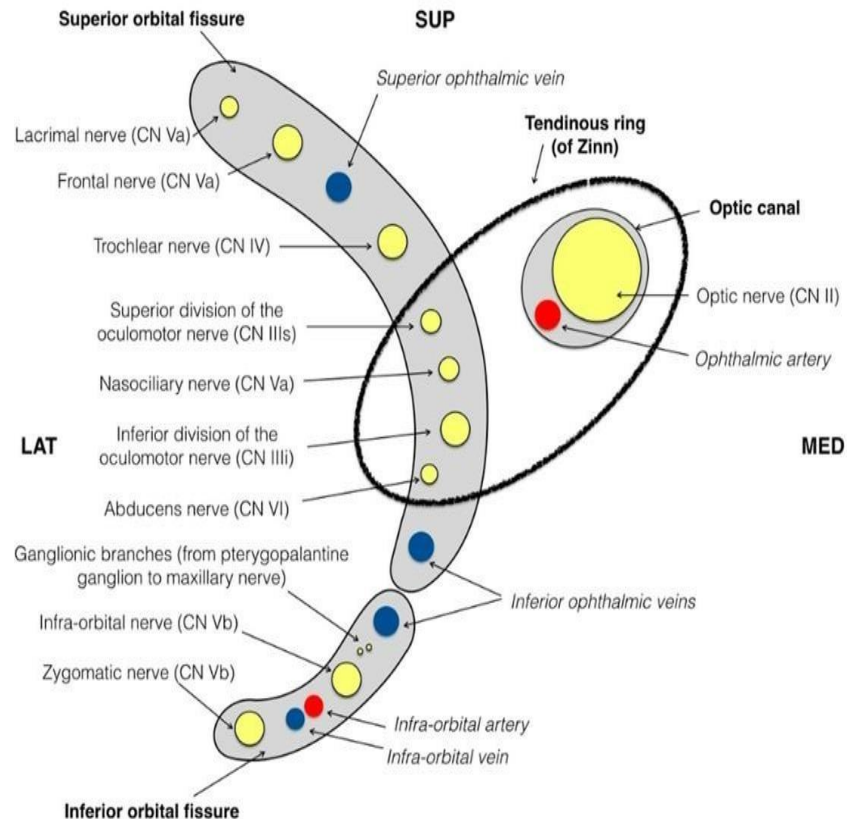
- The **orbit** is a four-sided **pyramidal** space with its **apex** located posteriorly at the optic foramen.
- Each cavity is essentially a **socket** for the **eyeball** and its associated structures.
- The **seven bones** that form the orbit are the frontal, zygomatic, sphenoid, palatine, ethmoid, lacrimal, and maxilla.
- The **medial wall** of the two orbits is parallel to the midsagittal plane. However, the **lateral wall** makes a 45-degree angle with the medial wall at the orbital apex.
- The **apex** of the orbit:
 - The apex of the orbit is at the **optic foramen**.
 - It transmits **optic nerve & ophthalmic artery**.
- The **base** of the orbit:



- is called the **orbital margin** is the **external aperture of the orbit**. It is quadrilateral in shape and has **4 borders**:
- The supra-orbital, infra-orbital, lateral & medial margins.
- The **roof** of the orbit:
 - It is formed of **orbital plate** of frontal bone and **lesser wing** of sphenoid bone.
 - It separates the **orbit** from the **anterior cranial fossa**.
 - The **fossa for the lacrimal gland** is located at the anterolateral part of the roof.
 - The **trochlea** is the only cartilage found in the orbit through which the **tendon** of the **superior oblique muscle** passes.
- The **floor** of the orbit:
 - The **infraorbital groove**, which leads to the **infraorbital canal** for the **infraorbital neurovascular bundle**, which ends anteriorly at the **infraorbital foramen** on the face.
 - The floor separates the **orbit** from the **maxillary sinus**.
- The **lateral wall** of the orbit:
 - It separates the orbit from the temporal region and middle cranial fossa.
 - It is the strongest (thickest) wall of the orbit.
- The **medial wall** of the orbit:
 - It contains the anterior and posterior ethmoidal foramina which transmit the anterior and posterior ethmoidal vessels and nerves.
 - **Important landmarks** on the medial wall are:
 - 1- The anterior lacrimal crest gives attachment to medial palpebral ligament.
 - 2- The posterior lacrimal crest gives attachment to lacrimal part of orbicularis oculi muscle.
 - 3- The lacrimal groove is a depression between anterior and posterior lacrimal crests. It lodges **lacrimal sac**. It communicates inferiorly to the nasolacrimal canal which ends in the **inferior meatus of nasal cavity**.
- The **superior orbital fissure**:
 - It communicates with the middle cranial fossa.
 - Its central part gives

attachment of **common tendinous ring (anulus of Zinn)**.

- It transmits 3rd, 4th, 6th cranial nerves, (frontal, lacrimal and Naso ciliary nerve of ophthalmic division of trigeminal nerve, Superior & inferior ophthalmic veins, and Orbital branch of middle meningeal artery.



▪ The **inferior orbital fissure**:

- It communicates with both the pterygopalatine fossa and the infratemporal fossa.
- It transmits **branches of the Infraorbital artery, Infraorbital nerve, Zygomatic nerve, Communicating veins between (the pterygoid plexus of veins and the inferior ophthalmic vein), and orbital branches of pterygopalatine ganglion.**

Contents of the orbit:

▪ **Periorbital Fascia and fat**

▪ **Extraocular Muscles:**

- There are **seven** voluntary muscles in the orbit. **Six** are attached to and move the eyeball while the **seventh** elevates the upper eyelid. There are smooth muscle fibers in the eyelids.
- The six skeletal muscles that attach to the eyeball are the four rectus ("straight") muscles and the two oblique muscles. Each muscle inserts into the sclera behind the cornea.

➡ **Four recti muscles (Superior, Inferior, Medial and Lateral recti muscles):**

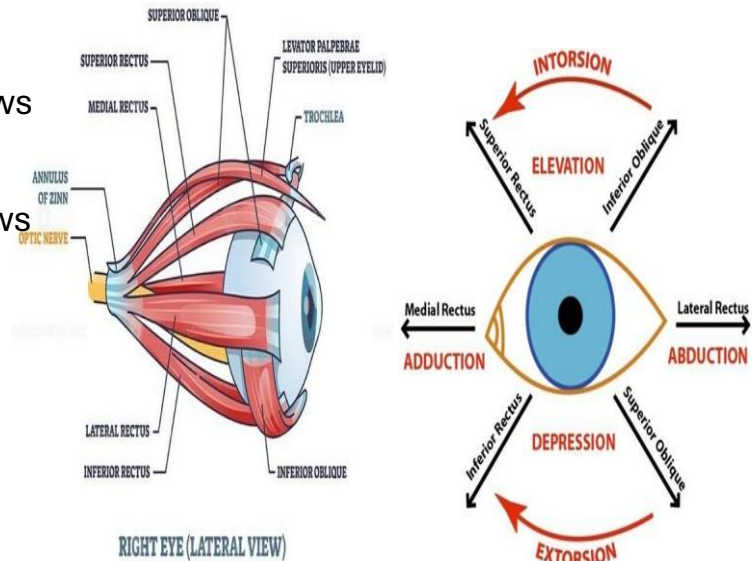
- **Origin:** They arise from the **common tendinous ring of extraocular muscles (anulus of Zinni)** that surrounds the optic foramen and the middle

portion of the superior orbital fissure. It is derived from the periorbital and is actually **oval in shape**.

- **Insertion:** are inserted into the **sclera** at varying distances from the corneoscleral junction.
- **Nerve supply:** all are supplied by **oculomotor nerve** except the **lateral rectus muscle** is supplied by **abducent nerve**.

- **Action:**

- The **medial rectus muscle** allows **eye adduction**.
- The **lateral rectus muscle** allows **eye abduction**.
- The **superior rectus muscle** is responsible for **eye elevation, intorsion and adduction**.
- The **inferior rectus muscle** is responsible for **eye depression, extorsion and adduction**.



■ **Levator palpebrae superioris muscle:**

- **Origin:** arises from the inferior surface of the lesser wing of the sphenoid bone, above the superior rectus muscle.
- **Insertion:** It has “multiple” insertions, include the following: insert into the superior tarsus, the superior conjunctival fornix and the superficial fascia of the upper eyelid.
- The **middle portion** of this muscle contains smooth muscle fibers, **Müller’s muscle**. Which is responsible for the tone of the eyelid and is innervated by **sympathetic fibers**.
- **Nerve supply:** superior division of **oculomotor nerve**.
- **Action:** elevates the upper eyelid and pulls upwards the superior conjunctival fornix.

■ **Superior Oblique Muscle:** It is the longest and most slender muscle.

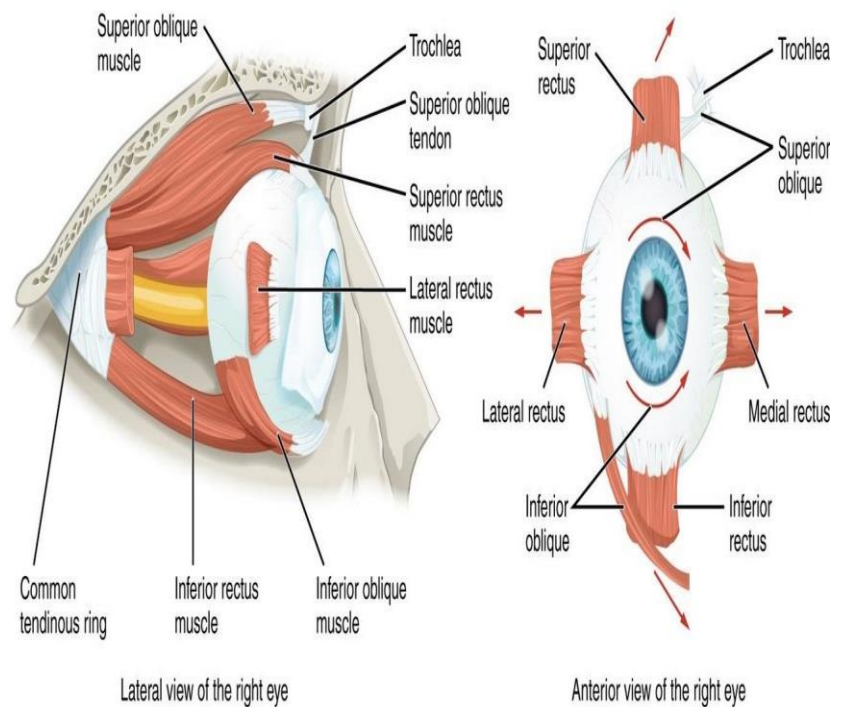
- It is found at the superior and medial side of the orbit.

- **Origin:** It arises from the body of the sphenoid bone.

- **Insertion:** It forms a rounded tendon that passes through a fibrocartilaginous (trochlea). This tendon then passes laterally and posteriorly, running beneath the superior rectus muscle to insert into the superior lateral quadrant of the sclera posterior to equator.

- **Nerve supply:** Trochlear nerve.

- **Action:** Eye depression, intorsion, and abduction.



■ Inferior oblique muscle:

- It is the shortest of the extraocular muscles.
- **Origin:** It arises anteriorly from the **floor** of the orbital surface of maxilla just laterally to lacrimal groove.
- **Insertion:** The muscle runs laterally to get inserted into the **lateral inferior quadrant of the sclera**, posterior to **equator**, between superior and lateral recti muscles.
- **Nerve supply:** Oculomotor nerve.
- **Action:** Eye elevation, extorsion, and abduction.

Sensory nerves in Orbit:

1- The **optic nerve (CN II):**

- It is the sensory nerve of **vision**. It is about **4cm** long.
- It pierces the sclera **4mm** medial to the center of back of eyeball.
- Consists of the **axons** of the **retinal ganglion cells**.
- The nerve takes a **sinuous course** which allows free movements for the eyeball. It traverses the **orbit** through the **optic canal** into the **middle cranial fossa** to join the **optic chiasma**.

2- Zygomatic nerve:

- It is a branch of maxillary nerve.
- It enters the orbit through the inferior orbital fissure.
- It is divided into zygomaticotemporal and zygomaticofacial branches which run through the corresponding canals.

3- Ophthalmic division of the trigeminal nerve (V1) is the sensory nerve to the orbit. It gives **3 branches** before passing through the superior orbital fissure:

a- Frontal nerve: It enters the orbit through the superior orbital fissure.

- The frontal nerve divides into **supratrochlear** and **supraorbital branches** supplying skin of forehead.

b- Lacrimal nerve: It enters the orbit through the superior orbital fissure.

- It is sensory to lacrimal gland.
- c- Nasociliary nerve:** It enters the orbit through the superior orbital fissure.
- It gives off long ciliary branches, posterior ethmoidal branch, infratrochlear branch, and ends as anterior ethmoidal nerve.
- It supplies the medial aspect of both eyelids, base of the nose, ethmoidal air cells, sphenoid sinus, meninges, cornea, sclera, iris, and ciliary body.

Motor nerves in the orbit: *they* enter the orbit through superior orbital fissure.

1-The oculomotor nerve:

- It is divided into **superior & inferior divisions** prior to entering the superior orbital fissure.
- Supplying all extraocular muscles **except superior oblique & lateral rectus muscles**.
- Contains preganglionic parasympathetic fibers that synapse in the ciliary ganglion and innervate the ciliary and sphincter pupillae muscles.

2- The trochlear nerve supplies **superior oblique muscle**.

3- The **abducent nerve** innervates **lateral rectus muscle**.

Ciliary ganglion:

- This is a minute **parasympathetic ganglion** which lies in between the lateral rectus and optic nerve.
- It is situated posterior to the globe and anterior to the superior orbital fissure.

Roots:

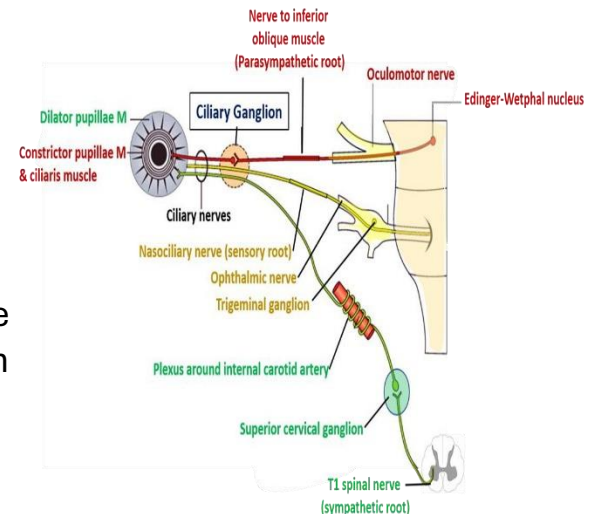
1- **Sensory root:** from **Naso ciliary nerve**.

2- **Sympathetic root:** from the plexus around internal carotid artery.

3- **Parasympathetic root:**

Preganglionic parasympathetic fibers arising from the **Edinger-Westphal nucleus**, passing with **oculomotor nerve** then with **nerve to inferior oblique**.

- **Branches:** these are short ciliary nerves supplying the **ciliary** and **sphincter pupillae muscles**.



Ophthalmic artery:

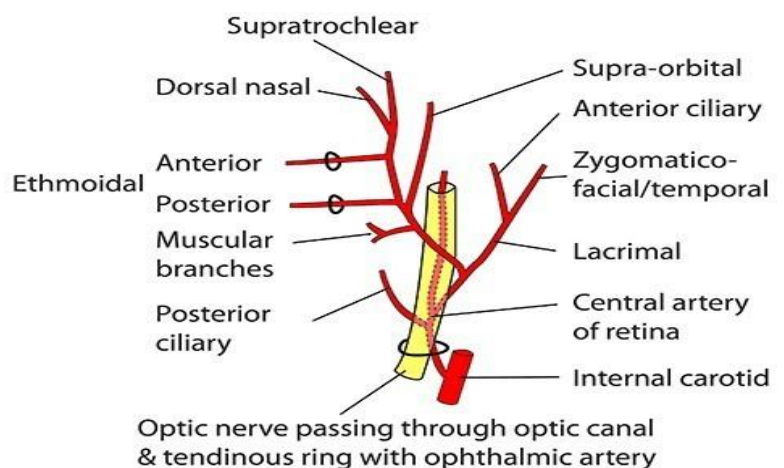
- arises from the **cerebral part** of **internal carotid artery**.
- It enters the orbit through the **optic foramen** inferior and lateral to the optic nerve.
- It divides into two terminal branches, **supratrochlear** and **dorsal nasal arteries**.
- **Branches:**

1- **Central artery of the retina** supplies the inner layers of the retina.

2- **Lacrimal artery** supplies lacrimal gland.

3- **Ciliary arteries** supply the **choroid** and outer layers of the **retina** (pigmented epithelium, rods, and cones).

4- **Supraorbital and supratrochlear arteries** supply scalp and face.



- 5- **Anterior and posterior ethmoidal arteries** to the nasal cavity, ethmoidal air sinuses and meninges of the anterior cranial fossa.

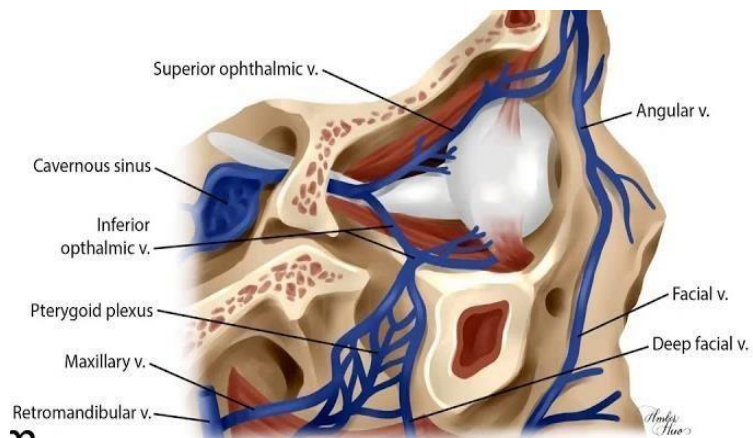
Ophthalmic Veins:

1- The superior ophthalmic vein:

- It begins behind the edial end of the upper eyelid.
- It connects with anterior facial vein.
- It receives tributaries corresponding to branches of ophthalmic artery.
- It passes through superior orbital fissure to empty into the **cavernoussinus**.

2- The inferior ophthalmic vein:

- It is formed by a plexus that lies on the orbital floor.
- It sends a communicating branch through inferior orbital fissure to anastomose with pterygoid plexus of veins.
- It passes through superior orbital fissure to empty into the **cavernous sinus**.



Eyeball:

- The eyeball is embedded in the orbital fat, but it is separated from it by fascial sheath of the eyeball (Tenon's capsule) forms a socket-like structure that helps suspend the eyeball
- It can be divided into **3 coats: fibrous, vascular and inner layers**.

1- **Fibrous (Outer) coat:** It consists of **cornea and sclera**. o

Cornea:

- is the **anterior transparent** part of the fibrous layer.
- It constitutes an anterior **1/6** of the fibrous layer.
- It is non-vascular and no

lymphatics. o **Sclera:**

- **Sclera** constitutes the posterior **5/6** of the globe.
- It is pierced posteriorly by **optic nerve** at **lamina cribrosa**.
- It provides attachment to the extraocular muscles.
- It is the white part of the eye.

2- Vascular (Middle) coat (Uveal tract):

It is composed of choroid, ciliary body & iris.

a) The **choroid**:

- The choroid constitutes the **greater posterior** part **5/6** of this coat.
- It is a thin, **pigmented highly vascular** connective tissue layer of the eyeball.
- It provides nourishment to the outer retina.

b) The **ciliary body**:

- It is located between the **choroid** posteriorly and the **iris** anteriorly.
- It consists of two parts: **the ciliary muscle and ciliary processes**.
- It is attached to the **lens** via **suspensory ligaments**. It controls the shape of the **lens and** contributes to the formation of **aqueous humor**.

c) The **iris**:

- is the circular-colored **part** of the eye.
- The central hole of the iris is the **pupil**.
- Within the iris there are smooth muscle fibers that control the size of the pupil. They form **sphincter pupillae muscle**, which constricts the pupil, and **dilator pupillae muscle**, which dilates the pupil.

3- **Neural (inner) coat**: It is composed of the **retina**, which is made of neural and pigmented layers:

- o The **Pigmented (outer) layer** – formed by a single layer of cells. It is attached to the choroid and supports the choroid in absorbing light. It continues around the whole inner surface of the eye.
- o The **neural (inner) layer**– is deficient in the anterior portion of the eye and is composed of **photoreceptors**. The Centre of the posterior aspect of the retina is called **macula lutea**, within which lies the **fovea centralis** that contains the highest concentration of **cones photoreceptors** in the retina and is responsible for **high acuity vision**. The area where the optic nerve fibers leave the retina is called the **optic disc**; this area contains no photoreceptors (**blind spot**).

Lens:

- The **lens** of the eye is a **biconvex, transparent**, and **avascular** body which is in front of the **vitreous humour** and behind the **pupil**.
- The shape of the lens is altered by the ciliary body.

Anterior and posterior chambers:

- These are **fluid-filled areas** within the eye.
- The **anterior chamber** is located between the cornea and the iris.
- The **posterior chamber** behind the iris.

Anterior and posterior eye segments:

- The part in front of the eye lens is the **anterior segment** (filled with aqueous humor).
- The posterior part to the eye lens is the **posterior segment** (filled with **Vitreous humor**). The vitreous body is the largest structure of the eyeball, occupying four-fifths of the entire eye.

The lacrimal apparatus:

- The **lacrimal apparatus** is to produce and drain tears.

Components of lacrimal apparatus:

- The **lacrimal gland**: Secretes the tears. It is formed of **two parts: larger orbital part** which is in **lacrimal fossa** in the anterolateral part of the roof of orbit. The **smaller palpebral part** is lodged in the lateral part of the **upper eyelid**.
- About 10 excretory ducts arise from the lacrimal gland and convey the tear to the surface of the human eye.
 - The **lacrimal canaliculi** of the upper and lower eyelids collect tears towards the lacrimal sac.
 - The **lacrimal sac** is in the **lacrimal groove**. Its lower end is continuous with the nasolacrimal duct.
 - The **nasolacrimal duct**, by which the fluid is conveyed into the **inferior meatus** of the nasal cavity.
- The **nerve supply** of the lacrimal apparatus involves both a **sympathetic plexus** around the internal carotid artery, and **parasympathetic** from the greater petrosal nerve of facial nerve (**CN VII**), which relays in **pterygopalatine ganglion**.

